# Nicolò Iacobone

Computer Vision & 3D Deep Learning Researcher

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## About Me

I am a computer vision researcher working across feed-forward 3D reconstruction and semantic scene understanding. I am looking for research or ML engineering roles in industry, or a PhD in 3D perception and representation learning. Outside engineering, I build things with Arduino, I juggle, fish and solve Rubik's cubes, and every season I reach Grand Champion in Clash Royale, which puts me among the top 2–4% of active players worldwide.

## Education

**MSc in Computer Engineering — Artificial Intelligence and Data Analytics** **Oct 2022 – Mar 2026**
*Politecnico di Torino, Italy* · 108/110 · Thesis carried out at ETH Zürich (Jul 2025 – Mar 2026)

**BSc in Computer Science and Automation Engineering** **Oct 2019 – Oct 2022**
*Politecnico di Bari, Italy* · Erasmus exchange at Universitatea Tehnică din Cluj-Napoca, Romania (Oct 2021 – Jul 2022)

## Research Experience

**Research Collaborator — ETH Zürich, Photogrammetry & Remote Sensing (D-BAUG)** **Mar 2026 – Sep 2026**
*3D-Consistent Instance Segmentation on Frozen Feed-Forward Backbones* · *Code*

- Taught a frozen 3D reconstruction model (VGGT) to also pick out the individual objects in a scene, keeping each object consistent across every view, while training only on ordinary 2D annotations.
- Built the whole pipeline myself, from data loading to evaluation on the official ScanNet benchmark, where it beats the two published methods it compares against and trains far faster.

**Master's Thesis — ETH Zürich** **Jul 2025 – Mar 2026**
*Towards Multi-View 3D Instance Segmentation via Semantic Distillation* · *Code* · *SAM 2 teacher fork*

- Taught a 3D reconstruction model (MapAnything) to also understand what it sees, by having it imitate the features of a segmentation model (SAM 2) while staying consistent across views.
- Found that a geometry-focused model runs out of capacity for fine-grained semantics, which led to the follow-up work above.

## Projects

**Independent work on open-source models** **Jul 2025 – Mar 2026**

- **Automatic 2D instance annotation of ScanNet with SAM 3** (code): a pipeline that labels every object in each ScanNet scene on its own, tracking objects across frames and picking them up again when they are lost.
- **Controlled synthetic 3D data and point tracking** (Kubric, SpaTrackerV2): generated synthetic scenes with controlled camera motion and ground-truth object masks, from static to cluttered and occlusion-heavy, and tracked 3D points through them.

**Academic projects**

- **Real-Time Anomaly Detection for Road Scenes:** real-time inference on embedded devices — Report · Jun 2024
- **Synthetic Data Generation for AROL Closure Systems:** CVAE for industrial sensor time series — Code & slides · Mar 2024
- **HoloLens 2 Mixed-Reality App:** Unity/MRTK holograms that follow the user's head and hands — Code · Jun 2022

## Skills

**3D Computer Vision:** 3D reconstruction & scene understanding · multi-view perception · 2D/3D instance segmentation · 3D point tracking

**Deep Learning:** Knowledge distillation · object detection & segmentation · representation learning · synthetic data generation

**Programming:** Python (advanced), C/C++, PyTorch, HPC / SLURM **Languages:** Italian (native) · English (proficient)